

What was inspected

97 natural eval-val proteins, exp277 epoch 2, step 479417. 100 saved rollout-and-resample predictions per protein; no new inference. All 1,940 original metric rows reproduced exactly. HTML atlas shows every predicted/true map, votes and errors, with zoom. No eval-test analysis.

The strongest visual pattern

Severe failures replace long-range traces with near-diagonal contact clusters. Worst accuracy quartile: 69.3% long-range truth vs 41.2% predictions. Examples: 8arl_A, 7wz5_A, 8bux_A, 8dmu_A. Better maps recover broad features but often miss detailed contacts. Positive control: 7y8h_A reaches 84.3% exact precision.

Why visual resemblance can mislead

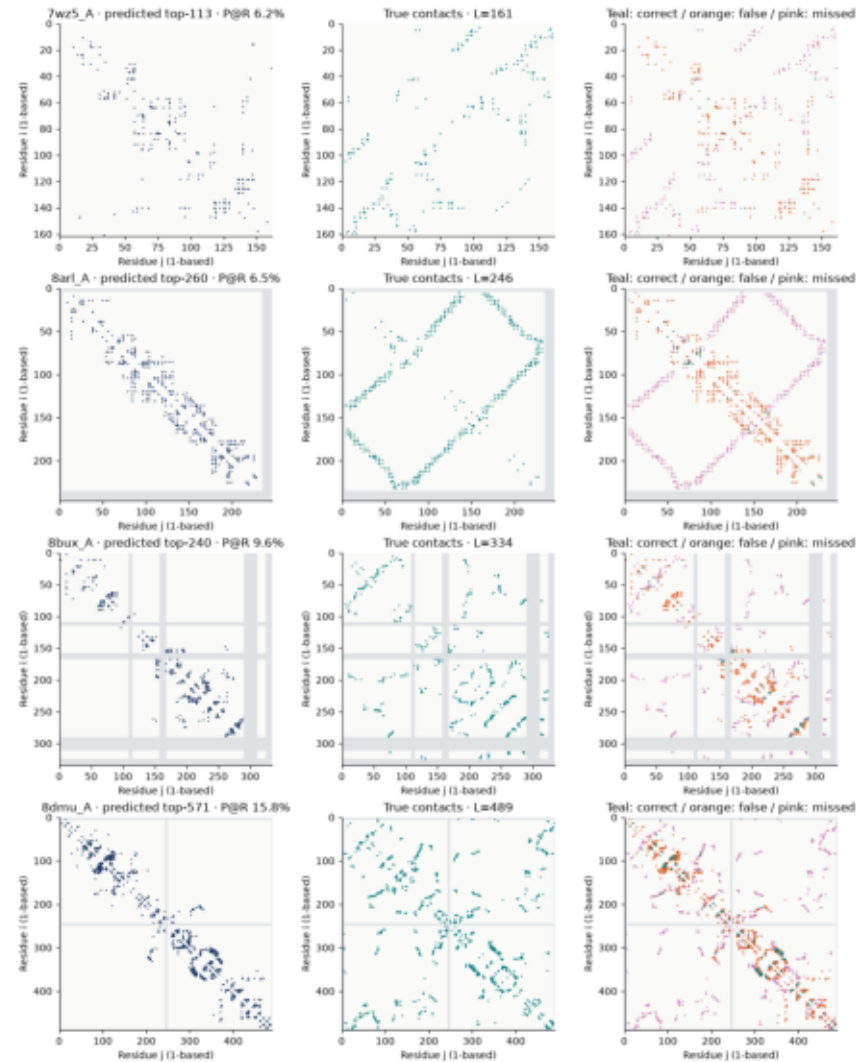
Exact precision: 55.75%. Many-to-one proximity within two residues: 80.72%. Keep exact matches and pair FP with FN one-to-one: only 64.01%. The truth-informed local correction gains 8.26 percentage points. Most errors remain; local proximity is not a substitute for exact accuracy.

Sampling support and next steps

93.9% of true contacts appear in at least one rollout. 29.3% get fewer than ten votes; top-L/5 precision is 81.2%. Correct contacts are often present but weakly ranked. Marginal votes cannot reveal within-rollout consistency or competing folds. Next: inspect whole-map samples and test truth-free structural selection. Training causes remain hypotheses; these maps do not establish one.

cases_1.png

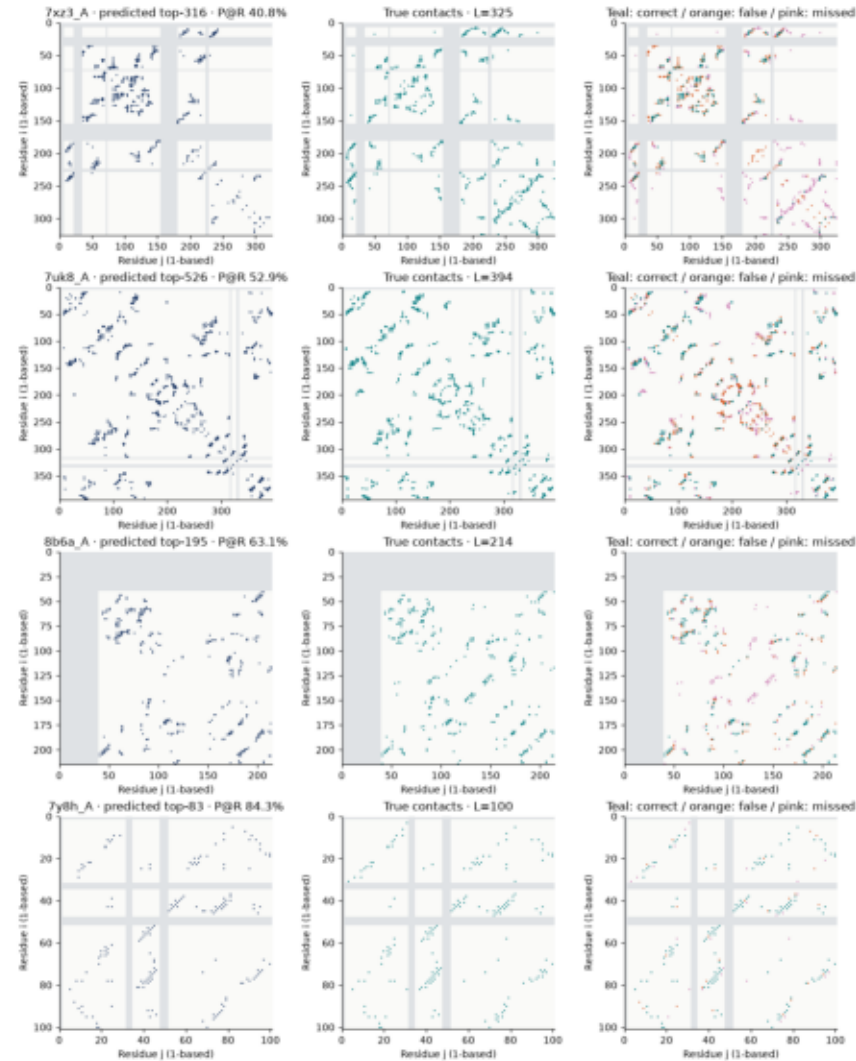
Selected exploratory cases: predicted top-R, truth, and errors. Both halves mirrored; gray bands are unresolved residues.



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$ plot_results.py
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cases_2.png

Selected exploratory cases: predicted top-R, truth, and errors. Both halves mirrored; gray bands are unresolved residues.



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$ plot_results.py
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diagnostics.png

All 97 eval-val proteins. Local correction uses a truth-informed maximum one-to-one matching, fixing exact matches first; it is an upper bound, not an improved predictor.

